Hyodae Seo

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BIO

Dr. Hyodae Seo is an Associate Professor in the Department of Oceanography and an Associate Director of the Uehiro Center for the Advancement of Oceanography. He is a physical oceanographer and climate scientist with a broad range of research and teaching interests in ocean and atmospheric processes and their coupled interactions, with a focus on weather and climate. He combines high-resolution coupled modeling with theories of geophysical fluid dynamics and analyses of in situ and satellite observations to better understand ocean-atmosphere boundary layer processes, improve their representation in numerical models, and evaluate their influence on the ocean, weather, and climate. Dr. Seo actively contributes to developing sustainable ocean-observing strategies and strives to engage the public on issues related to the ocean, extreme weather, climate, and ocean-based renewable energy research.

EDUCATION

2007: Ph.D. in Oceanography, Scripps Institution of Oceanography, UC San Diego
Thesis title: Mesoscale coupled ocean-atmosphere interaction
Thesis advisors: Dr. Arthur J. Miller and Dr. John Roads (deceased)
2002: B.S. in Meteorology, Yonsei University, Seoul, South Korea

POSITIONS HELD

2024 – : Associate Professor, Department of Oceanography, University of Hawaiʻi at Mānoa
2023 – 2024: Senior Scientist, Woods Hole Oceanographic Institution (WHOI)
2018 – 2023: Associate Scientist with Tenure, WHOI
2014 – 2018: Associate Scientist, WHOI
2010 – 2014: Assistant Scientist, WHOI
2009 – 2010: Visiting Assistant Researcher, IPRC, University of Hawaiʻi at Mānoa
2007 – 2008: NOAA Climate and Global Change Postdoctoral Fellow, UCLA & UH Mānoa
2007: Visiting Scientist, International Pacific Research Center, Univ. Hawaiʻi at Mānoa
2002 – 2007: Graduate Research Assistant, Scripps Institution of Oceanography, UCSD

AWARDS AND HONORS

- WHOI Francis E. Fowler IV Ocean and Climate Fellow (2022, link)
- Office of Naval Research (ONR) Young Investigator Award (2015, WHOI press release)
- NOAA Climate and Global Change Postdoctoral Fellowship (2007-2009, class 17)
- Physical Oceanography Dissertation Symposium V, Honolulu (2008)
- NCAR Advanced Study Program (ASP) Graduate Student Visiting Fellowship (2006)
- Scripps Institution of Oceanography Frieman Director's Prize for Excellence in Graduate Student Research (2006, UCSD press release)

- Outstanding Student Paper Award, Ocean Sciences Meeting, Honolulu (2006)

PUBLICATIONS (*led by student and postdoc under my direct supervision)

Under Review, Revision, or In Press:

- Guo, Y., Carolina Castillo-Trujillo, A., K. Chen, Y.-O. Kwon, S. Perkins, **H. Seo**, P. Fratantoni, M. Alexander, V. Saba, Multi-Year Predictability of Ocean Conditions on the U.S. Northeast Shelf Using Dynamical Downscaling. *J. Geophys. Res. Oceans, Submitted.*
- Song, H., H. Seunu, **H. Seo**, D. J. McGillicuddy, Jr., K. Kwak, J. Yun, H.-J. Kim, J. Lee, K. Park, and J. Park, Physical and Biogeochemical Characteristics of Four Types of Eddies in the Southern Ocean, *Geophys. Res. Lett*., *Submitted*
- Kirinchch, A., and coauthors, including **H. Seo**, and C. Sauvage, Improving the Understanding and Forecasting of Winds over the Northeast U.S. Shelf: The Third Wind Forecast Improvement Project (WFIP3). *Bull. Amer. Meteo. Soc*., *Submitted*
- *Renkl, C., **H. Seo**, and A. J. Miller, Marine heatwave impacts on landfalling atmospheric river on the U.S. West Coast. *Submitted*
- *Kim, Y.-J., K. Tanaka, **H. Seo**, K. Komatsum, Y. Matsumura, C. Sauvage, and C. Renkl, A Study on the Role of Time- and Spatially Varying Wind in Coastal Circulation Using High-resolution Oceanic and Atmospheric Numerical Simulations. *J. Geophys. Res. Oceans*, *Submitted*
- Renault, L., J. Wenegrat, **H. Seo**, P. Marchesiello, The Energetics of Fine-Scale Air-Sea Interactions. *Annual Review of Marine Science*, *In Press*

Published:

2026

69. Cho, A., H. Song, I.-J. Moon, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, A. J. Miller, 2026: Modulation of Tropical Cyclone Intensity by Current-Wind Interaction, *npj Clim. Atmos. Sci*., 9, 44, https://doi.org/10.1038/s41612-025-01316-1

2025

68. *Sauvage, C., **H. Seo**, S. Zippel, C.-A. Clayson, and J. B. Edson, 2025: Fetch-dependent Surface Wave Responses To Offshore Wind Farms in the Northeast U.S. Coast, *J. Geophys. Res. Oceans*, 130, e2025JC023156, https://doi.org/10.1029/2025JC023156
67. **Seo, H.**, C. Sauvage, C. Renkl, J. K. Lundquist, and A. Kirincich, 2025: Sea Surface Warming and Ocean-to-Atmosphere Feedback Driven by Large-Scale Offshore Wind Farms Under Seasonally Stratified Conditions. *Sci. Adv*., 11, eadw7603, https://www.science.org/doi/10.1126/sciadv.adw7603
66. Raj, A., B. P. Kumar, V. Jampana, P. G. Remya, **H. Seo**, N. Sureshkumar, and P. R. Rao, 2025: Wind-Stress Misalignment in the Presence of Swells in the Bay of Bengal. *Frontiers in Marine Science*, 12. https://doi.org/10.3389/fmars.2025.1647849
65. Yang, S., H. Bae, M.-S. Park, M. Bourassa, C. C. Nam, S. Cocke, D. W. Shin, B. W. Barr, **H. Se**o, D.-H. Cha, M.-H. Kwon, D. Kim, K.-Y. Jung, and B.-M. Kim, 2025: Sea spray effects on typhoon prediction in the Yellow and East China Seas: Case studies using a coupled atmosphere-ocean-wave model for Lingling (2019) and Maysak (2020). *Env. Res. Lett*., 20, 054028. https://doi.org/10.1088/1748-9326/adc616
64. Cho, A., H. Song, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, B. D. Cornuelle, and A. J. Miller, 2025: Dynamic and Thermodynamic coupling between the Atmosphere and Ocean near the Kuroshio Current and Extension System. *Ocean Modell*., 194, 102496. https://doi.org/10.1016/j.ocemod.2024.10249

2024

63. *Sauvage, C., **H. Seo**, B. W. Barr, J. B. Edison, and C. A. Clayson, 2024: Misaligned Wind-Waves Behind Atmospheric Cold Fronts. *J. Geophys. Res. Oceans*, 129, e2024JC021162. https://doi.org/10.1029/2024JC021162
62. Shinoda, T., T. G. Jensen, Z. Lachkar, Y. Masumoto, and **H. Seo**, 2024: Modeling the Indian Ocean, Intraseasonal variability in the Indian Ocean region. The Indian Ocean and its role in the global climate system. Elsevier Publishing, Editors: C. C. Ummenhofer and R. R. Hood. https://doi.org/10.1016/B978-0-12-822698-8.00013

2023

61. *Steffen, J. D., **H. Seo**, C. A. Clayson, S. Pei, and T. Shionda, 2023: Impacts of Tidal Mixing on Diurnal to Intraseasonal Air-Sea Interactions in the Maritime Continent. *Deep-Sea Res.-II*, 212, 105343. https://doi.org/10.1016/j.dsr2.2023.105343
60. *Castillo-Trujillo A. C., Y.-O. Kwon, P. Fratantoni, K. Chen, **H. Seo**, M. A. Alexander, and V. S. Saba, 2023: An evaluation of eight global ocean reanalyses for the Northeast U.S. continental shelf. *Prog. Oceanogr*., https://doi.org/10.1016/j.pocean.2023.103126
59. Clayson, C. A., C. A. DeMott, S. P. de Szoeke, P. Chang, G. R. Foltz, R. Krishnamurthy, T. Lee, A. Molod, D. G. Ortiz-Suslow, J. Pullen, D. H. Richter, **H. Seo**, P. C. Taylor, E. Thompson, B. V. Bôas, C. J. Zappa, and P. Zuidema, 2023: A New Paradigm for Observing and Modeling of Air-Sea Interactions to Advance Earth System Prediction. A US CLIVAR Report, US CLIVAR Project Office, 86 pp. https://doi.org/10.5065/24j7-w583.
58. Beaudin, E., E. Di Lorenzo, A. J. Miller, **H. Seo**, and Y. Joh, 2023: Impact of Extratropical Northeast Pacific SST on U.S. West Coast Precipitation, *Geophys. Res. Lett*., 50, e2022GL102354. https://doi.org/10.1029/2022GL102354
57. *Sauvage, C., **H. Seo**, C. A. Clayson, and J. B. Edson, 2023: Improving wave-based air-sea momentum flux parameterization in mixed seas. *J. Geophys. Res. Oceans*, 128, e2022JC019277. https://doi.org/10.1029/2022JC019277
56. **Seo, H.**, L. W. O'Neill, M. A. Bourassa, A. Czaja, K. Drushka, B. Fox-Kemper, I. Frenger, S. T. Gille, B. P. Kirtman, S. Minobe, A. G. Pendergrass, L. Renault, M. J. Roberts, N. Schneider, R. J. Small, A. Stoffelen, and Q. Wang, 2023: Ocean Mesoscale and Frontal-scale Ocean-Atmosphere Interactions and Influence on Large-scale Climate: A Review. *J. Climate*, 36, 1981-2013. https://doi.org/10.1175/JCLI-D-21-0982.1

2021

55. Kwak, K. H. Song, J. Marshall, **H. Seo,** and D. McGillicuddy, Jr., 2021: Suppressed pCO2 in the Southern Ocean due to the interaction between current and wind. *J. Geophys. Res. Oceans*, 126, e2021JC017884. https://doi.org/10.1029/2021JC01788
54. Shinoda, T., S. Pei, W. Wang, J. X. Fu, R.-C. Lien, **H. Seo**, and A. Soloviev, 2021: Climate Process Team: improvement of ocean component of NOAA Climate Forecast System relevant to Madden-Julian Oscillation simulations. *J. Adv. Model. Earth Syst.*, 13, e2021MS002658. https://doi.org/10.1029/2021MS002658
53. **Seo, H.**, H. Song, L. W. O'Neill, M. R. Mazloff, and B. D. Cornuelle, 2021: Impacts of ocean currents on the South Indian Ocean extratropical storm track through the relative wind effect. *J. Climate*, 34, 9093-9113. https://doi.org/10.1175/JCLI-D-21-0142
52. Pei, S., T. Shinoda, J. Steffen, and **H. Seo**, 2021: Substantial Sea Surface Temperature cooling in the Banda Sea associated with the Madden-Julian Oscillation in the boreal winter of 2015. *J. Geophys. Res. Oceans*., 126, e2021JC017226. https://doi.org/10.1029/2021JC017226

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51. Shroyer, E., and Co-authors, including **H. Seo**, 2021: Bay of Bengal Intraseasonal Oscillation and the 2018 Monsoon Onset. Bull. Amer. Meteo. Soc., 102(10), E1936-E1951. https://doi.org/10.1175/BAMS-D-20-0113
50. *Bartusek, S., **H. Seo**, C. C. Ummenhofer, and J. D. Steffen, 2021: The role of nearshore air-sea interactions for landfalling atmospheric rivers on the U.S. West Coast. *Geophys. Res. Lett.*, 48, e2020GL091388. https://doi.org/10.1029/2020GL091388
49. Sun, R., A. C. Subramanian, B. D. Cornuelle, M. R. Mazloff, A. J. Miller, F. M. Ralph, **H. Seo**, and I. Hoteit, 2021: The role of air-sea interactions in atmospheric river events: Case studies using the SKRIPS regional coupled model. *J. Geophys. Res. Atmosphere*, 126, e2020JD032885. https://doi.org/10.1029/2020JD032885

2020

48. Liang, Y.-C., M.-H. Lo, C.-W. Lan, **H. Seo,** C. C. Ummenhofer, S. Yeager, R.-J. Wu, and J. D. Steffen, 2020: Amplified Seasonal Cycle in Hydroclimate over the Amazon River Basin and its Plume Region in the Atlantic. *Nat. Commun*., 11, 4390. https://doi.org/10.1038/s41467-020-18187-0
47. *Yuan, X., C. C. Ummenhofer, **H. Seo**, and Z. Su, 2020: Relative contributions of heat flux and wind stress on the spatiotemporal upper-ocean variability in the tropical Indian Ocean. *Env. Res. Lett*., 15, 084047. https://doi.org/10.1088/1748-9326/ab9f7f
46. Gentemann, C., C. A. Clayson, S. Brown, T. Lee, R. Parfitt, J. T. Farrar, M. Bourassa, P. J. Minnett, H. Seo, S. T. Gille, and V. Zlotnicki, 2020: FluxSat: Measuring the ocean-atmosphere turbulent exchange of heat and moisture from space. *Remote Sensing*, 12, 1796. https://doi.org/10.3390/rs12111796
45. Sprintall, J., K. A. Reed, A. H. Butler, G. R. Foltz, S. G. Penny, and **H. Seo**, 2020: Best Practice Strategies for Process Studies designed to Improve Climate Modeling, *Bull. Amer. Met. Soc.,* 101, E1842-E1850. https://doi.org/10.1175/BAMS-D-19-0263.
44. Song, H., J. Marshall, D. J. McGillicuddy Jr., and **H. Seo**, 2020: The impact of the current-wind interaction on the vertical processes in the Southern Ocean*. J. Geophys. Res.-Oceans*, 125, e2020JC016046. https://doi.org/10.1029/2020JC01604
43. Gopalakrishnan, G., A. C. Subramanian, A. J. Miller, **H. Seo**, and D. Sengupta, 2020: Estimation and prediction of the upper ocean circulation in the Bay of Bengal, *Deep-Sea Res. II,* 172, 104721. https://doi.org/10.1016/j.dsr2.2019.10472
42. Hagos, S., G. R. Foltz, C. Zhang, E. Thompson, **H. Seo**, S. Chen, A. Capotondi, K. A. Reed, C. DeMott, and A. Protat, 2020: Atmospheric Convection and Air-Sea Interactions over the Tropical Oceans: Scientific Progress, Challenges and Opportunities. *Bull. Amer. Met. Soc., 101*, E253–E258. https://doi.org/10.1175/BAMS-D-19-026
41. Kwon, Y.-O., **H. Seo**, C. C. Ummenhofer, and T. M. Joyce, 2020: Impact of Multidecadal Variability in Atlantic SST on Winter Atmospheric Blocking. *J. Climate*, 33, 867-892. https://doi.org/10.1175/JCLI-D-19-0324

2019

40. **Seo, H.**, A. C. Subramanian, H. Song, and J. S. Chowdary, 2019: Coupled effects of ocean current on wind stress in the Bay of Bengal: Eddy energetics and upper ocean stratification. *Deep-Sea Res. II*, 168, 104617. https://doi.org/10.1016/j.dsr2.2018.09.007
39. *Prend, C. J., **H. Seo**, R. A. Weller, and J. T. Farrar, 2019: Impact of freshwater plumes on intraseasonal upper ocean variability in the Bay of Bengal. *Deep-Sea Res.-II*, 161, 63-71. https://doi.org/10.1029/2018GL0810

38. Joyce, T. M., Y.-O. Kwon, **H. Seo**, and C. C. Ummenhofer, 2019: Meridional Gulf Stream shifts can influence wintertime variability in the North Atlantic Storm Track and Greenland Blocking. *Geophys. Res. Lett*., 46, 1702-1708. https://doi.org/10.1175/JCLI-D-18-0413.1
37. Weller, R. A., J. T. Farrar, **H. Seo**, C. J. Prend*, D. Sengupta, J. Sree Lekha, M. Ravichandran, and R. Venkatesen, 2019: Moored observations of the surface meteorology and air-sea fluxes in the northern Bay of Bengal in 2015, *J. Climate,* 32, 549-573. https://doi.org/10.1016/j.dsr2.2019.07.00

2018

36. *Parfitt, R., and **H. Seo**, 2018: A New Framework for Near-Surface Wind Convergence over the Kuroshio Extension and Gulf Stream in Wintertime: The Role of Atmospheric Fronts. *Geophys. Res. Lett*., 45, 9909–9918. https://doi.org/10.1175/JCLI-D-18-0413.1
35. Pullen, J., R. Allard, **H. Seo**, A. J. Miller, S. Chen, L. P. Pezzi, T. Smith, P. Chu, J. Alves, and R. Caldeira, 2018: Coupled ocean-atmosphere forecasting at short and medium time scales. *J. Mar. Res.,* 75, 877-921. https://elischolar.library.yale.edu/journal_of_marine_research/4
34. *Jin, X, Y.-O. Kwon, C. C. Ummenhofer, **H. Seo**, Y. Kosaka, and J. S. Wright, 2018: Distinct mechanisms of decadal subsurface heat content variations in the eastern and western Indian Ocean modulated by tropical Pacific SST, *J. Climate*, 31, 7751-7769. https://doi.org/10.1175/JCLI-D-18-0184.1
33. *Jin, X., Y.-O. Kwon, C. C. Ummenhofer, **H. Seo**, F. U. Schwarzkopf, A. Biastoch, C. W. Böning, and J. S. Wright, 2018: Influences of Pacific Climate Variability on Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean. *J. Climate*, 31, 4157-4174. https://doi.org/10.1175/JCLI-D-17-0654.1
32. Kwon, Y.-O., A. Camacho, C. Martinez-Zayas, and **H. Seo**, 2018: North Atlantic Eddy-driven Jet and Blocking Variability in the Community Earth System Model Version 1 Large Ensemble Simulations. *Climate Dyn.*, 51, 3275-3289. https://doi.org/10.1007/s00382-018-4078-6

2017

31. **Seo, H.**, Y.-O. Kwon, T. M. Joyce, and C. C. Ummenhofer, 2017: On the predominant nonlinear response of the extratropical atmosphere to meridional shift of the Gulf Stream. *J. Climate*, 30, 9679-9702. https://doi.org/10.1175/JCLI-D-16-0707.1
30. Morrow, R., L.-L. Fu, T. Farrar, **H. Seo**, P.-Y. Le Traon, 2017: Ocean eddies and mesoscale variability. Satellite Altimetry Over Oceans and Land Surfaces, In D. Stammer and A. Cazenave, editors, Satellite Altimetry Over Ocean and Land Surfaces. CRC Press, Taylor and Francis Group, Satellite Altimetry Over Oceans and Land Surfaces.
29. Centurioni, L. R., and Co-authors including **H. Seo,** 2017: Northern Arabian Sea Circulation-Autonomous Research (NASCar): A Research Initiative Based on Autonomous Sensors. *Oceanography*, 30, 74-87. https://doi.org/10.5670/oceanog.2017.224
28. Ummenhofer, C. C., **H. Seo**, Y-O. Kwon, R. Parfitt, S. Brands, and T. M. Joyce, 2017: Emerging European winter precipitation pattern linked to atmospheric circulation changes over the North Atlantic region in recent decades. *Geophys. Res. Lett*., 44, 8557–8566. https://doi.org/10.1002/2017GL074188
27. **Seo, H.**, 2017: Distinct influence of air-sea interactions mediated by mesoscale sea surface temperature and surface current in the Arabian Sea. *J. Climate*, 30, 8061-8079. https://doi.org/10.1175/JCLI-D-16-0834.1

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26. Miller, A. J., M. Collins, S. Gualdi, T. G. Jensen, V. Misra, L. P. Pezzi, D. W. Pierce, D. Putrasahan, **H. Seo,** and Y.-H. Tseng, 2017: Coupled Ocean-Atmosphere-Hydrology Modeling and Predictions, *J. Mar. Res*., 75, 361-402. https://elischolar.library.yale.edu/journal_of_marine_research/437
25. *Parfitt, R., A. Czaja, and **H. Seo**, 2017: A simple diagnostic for the detection of atmospheric fronts, *Geophys. Res. Lett*., 44, 4351–4358. https://doi.org/10.1002/2017GL073662
24. Jongaramrungruang, S., **H. Seo**, and C. C. Ummenhofer, 2017: Intraseasonal rainfall variability in the Bay of Bengal during the Summer Monsoon: Coupling with the ocean and modulation by the Indian Ocean Dipole. *Atmos. Sci. Lett*., 18, 88–95. https://doi.org/10.1002/asl.729

2016

23. Chowdary, J. S., G. Srinivas, T. S. Fousiya, A. Parekh, C. Gnanaseelan, **H. Seo**, and J. A. MacKinnon, 2016: Representation of Bay of Bengal Upper-Ocean Salinity in General Circulation Models. *Oceanography*, 29, 38–49. https://doi.org/10.5670/oceanog.2016.37
22. **Seo, H.**, A. J. Miller, and J. R. Norris, 2016: Eddy-wind interaction in the California Current System: dynamics and impacts, *J. Phys. Oceanogr*., 46, 439-459. https://doi.org/10.1175/JPO-D-15-0086.1
21. Brink, K. H. and **H. Seo**, 2016: Continental Shelf Baroclinic Instability 2: Oscillating wind forcing, *J. Phys. Oceanogr.*, 46, 569-582. https://doi.org/10.1175/JPO-D-15-0048.1

2015

20. *Oltmanns, M., F. Straneo, **H. Seo**, and G. W. K. Moore, 2015: The role of wave dynamics and small-scale topography for downslope wind events in southeast Greenland. J*. Atmos. Sci.*, 72, 2786-2805. https://doi.org/10.1175/JAS-D-14-0257.1
19. Cavanaugh, N., T. Allen, A. Subramanian, B. Mapes, **H. Seo**, A. J. Miller, 2015: The Skill of Tropical Linear Inverse Models in Hindcasting the Madden-Julian Oscillation. *Climate Dyn*., 44, 897-906. https://doi.org/10.1007/s00382-014-2181-x

2014

18. **Seo, H.**, A. C. Subramanian, A. J. Miller, and N. R. Cavanaugh, 2014: Coupled impacts of the diurnal cycle of sea surface temperature on the Madden-Julian Oscillation. *J. Climate*, 27, 8422-8443. https://doi.org/10.1175/JCLI-D-14-00141.1
17. Park, J.-Y., J.-S. Kug, **H. Seo**, and J. Bader, 2014: Impact of bio-physical feedbacks on the tropical climate in coupled and uncoupled GCMs. Climate Dyn., 43, 1811-1827. https://doi.org/10.1007/s00382-013-2009-0
16. **Seo, H**., Y.-O. Kwon, and J.-J. Park, 2014: On the effect of the East/Japan Sea SST variability on the North Pacific atmospheric circulation in a regional climate model. *J. Geophys. Res.-Atmospheres*, 119, 418-444. https://doi.org/10.1002/2013JD020523
15. Subramanian, A, M. Jochum, A. J. Miller, R. Neale, **H. Seo**, D. Waliser, and R. Murtugudde, 2014: The MJO and Global warming: A study in CCSM4. *Climate Dyn*., 42, 2019-2031. https://doi.org/10.1007/s00382-013-1846-1

2013

14. **Seo, H.,** and J. Yang, 2013: Dynamical response of the Arctic atmospheric boundary layer process to uncertainties in sea ice concentration. *J. Geophys. Res-Atmosphere*, 118, 12383-12402. https://doi.org/10.1002/2013JD020312
13. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Regional coupled ocean-atmosphere downscaling in the Southeast Pacific: Impacts on upwelling, mesoscale air-sea fluxes,

and ocean eddies. *Ocean Dynm*., 63, 463-488. https://doi.org/10.1007/s10236-013-0608-2

12. **Seo, H.** and S.-P. Xie, 2013: Impact of ocean warm layer thickness on the intensity of Hurricane Katrina in a regional coupled model. *Meteor. Atmos. Phys*., 122, 19-32. https://doi.org/10.1007/s00703-013-0275-3
11. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Isolating Mesoscale Coupled Ocean-Atmosphere Interactions in the Kuroshio Extension Region. *Dyn. Atmos. Oceans*, 63, 60-78. https://doi.org/10.1016/j.dynatmoce.2013.04.001

2012

10**. Seo, H**., K. H. Brink, C. E. Dorman, D. Koracin, and C. A. Edwards, 2012: What determines the spatial pattern in summer upwelling trends on the U.S. West Coast? *J. Geophys. Res.-Oceans,* 117, C08012. https://doi.org/10.1029/2012JC008016

9. Alexander, M.A., **H. Seo**, S.-P. Xie, and J.D. Scott, 2012: ENSO's impact on the gap wind regions of the eastern tropical Pacific Ocean. *J. Climate*, 25, 3549-3565. https://doi.org/10.1175/JCLI-D-11-00320.1

2011

8. S**eo, H**., and S.-P. Xie, 2011: Response and Impact of Equatorial Ocean Dynamics and Tropical Instability Waves in the Tropical Atlantic under Global Warming: A regional coupled downscaling study. *J. Geophys. Res.-Oceans,* 116, C03026. https://doi.org/10.1029/2010JC006670

2009

7. **Seo, H**., S.-P. Xie, R. Murtugudde, M. Jochum, and A. J. Miller, 2009: Seasonal effects of Indian Ocean freshwater forcing in a regional coupled model. *J. Climate*, 22, 6577-6596. https://doi.org/10.1175/2009JCLI2990

2008

6. **Seo, H.**, R. Murtugudde, M. Jochum, and A. J. Miller, 2008: Modeling of Mesoscale Coupled Ocean-Atmosphere Interaction and its Feedback to Ocean in the Western Arabian Sea. *Ocean Modell*., 25,120-131. https://doi.org/10.1016/j.ocemod.2008.07.003
5. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2008: Precipitation from African Easterly Waves in a Coupled Model of the Tropical Atlantic. *J. Climat*e, 21, 1417-1431. https://doi.org/10.1175/2007JCLI1906.1
4. Small, R. J., S. de Szoeke, S. P. Xie, L. O'Neill, **H. Seo**, Q. Song, P. Cornillon, M. Spall, and S. Minobe, 2008: Air-Sea Interaction over Ocean Fronts and Eddies. *Dyn. Atmos. Oceans*, 45, 274-319. https://doi.org/10.1016/j.dynatmoce.2008.01.001

2007

3. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2007: Feedback of Tropical Instability Wave - induced Atmospheric Variability onto the Ocean. *J. Climate*, 20, 5842-5855. https://doi.org/10.1175/JCLI4330.1
2. **Seo, H.**, A. J. Miller and J. O. Roads, 2007: The Scripps Coupled Ocean-Atmosphere Regional (SCOAR) model, with applications in the eastern Pacific sector. *J. Climate*, 20, 381-402. https://doi.org/10.1175/JCLI4016.1

2006

1. **Seo, H.**, M. Jochum, R. Murtugudde and A. J. Miller, 2006: Effect of Ocean Mesoscale Variability on the Mean State of Tropical Atlantic Climate. *Geophys. Res. Lett*., 33, L09606. https://doi.org/10.1029/2005GL025651

TEACHING

@ UHM (since August 2024)

- Spring 2026: OCN/ATMO 665: Small-Scale Air-Sea Interactions
- Spring 2025, OCN 760: Topics in Physical Oceanography (Air-Sea Interaction and Offshore Wind)
- Spring 2025, OCN780: Ocean Seminar
- Fall 2024, GES 311: The Changing Earth and Climate System. Guest lecture, "Why offshore wind energy: Necessity, Challenges and Oceanographers' Role", Nov 21

@ MIT/WHOI Joint Program (until July 2024)

- Spring Semester, 2023: Air-Sea Interaction and Boundary Layers, MIT/WHOI Joint Program Course 12.870. Co-teaching with Tom Farrar
- Fall Semester, 2017, 2019, 2021: Climate Variability and Diagnostics, MIT/WHOI Joint Program Course 12.860. Co-teaching with Caroline Ummenhofer
- Fall Semester, 2016, 2017, 2018, 2019: Introduction to (Observational) Physical Oceanography, MIT/WHOI Joint Program Course 12.808. Co-taught with John Toole

ADVISING AND MENTORING

Thesis Committee

@ UHM (since August 2024)

- M.S. Yuta Norden (University of Hawai'i, Oceanography), 2026- Present
- Ph.D. Meghan Bradley (University of Hawai'i, Oceanography), 2025- Present
- Ph.D. Ryo Dobashi (University of Hawai'i, Oceanography), 2025 — 2026
- Ph.D. Ajin Cho (Yonsei University), 2024— 2026
- Comprehensive Exam Committee, Carla Balzeau, Oceanography, UHM (2025)
- Comprehensive Exam Committee, Vera Stockmayer, Oceanography, UHM (2025)

@ MIT/WHOI Joint Program (until July 2024)

- Ph.D. General Exam Advisor for Anthony Meza (MIT-WHOI Joint Program), 2022-2023
- Ph.D. Thesis Committee for Hanyuan Liu (MIT-WHOI Joint Program), 2021-2023
- Ph.D. External Examiner for Christoph Renkl (Dalhousie University, Oceanography), 2020
- Ph.D. Thesis committee for Jared Buckley (Univ. Massachusetts, Dartmouth), 2015-2019
- M.S. Thesis committee for Xin Zhou (Florida Institute of Technology), 2015-2016
- M.S. Thesis committee for Sara Bosshart (MIT-WHOI Joint Program), 2013

Graduate Advisor

@ UHM (since August 2024)

- Rachel Wyn Pauly: Ph.D. student, Oceanography Department, UHM (2025–present)
- Clint Reyes: Summer graduate student, ORE, UHM (2025)

Research Supervisor

@ UHM (since August 2024)

- Dr. César Sauvage: Oceanographic Researcher (2024/11–present)
- Dr. Sid Kerhalkhar: Postdoc (2025/09–present)
- Allison Ho (2026–)

@ WHOI (until July 2024)

- Shikhar Rai (2023–)
- Benjamin Barr (2023–)
- Christoph Renkl (2023-2025) at WHOI. Now an Assistant Processor, University of Bonn
- Alma Carolina Castillo-Trujillo (2021-2024).
- César Sauvage (2020-2023), now RA3 at WHOI and UH
- John Steffen (2019-2021, now at NOAA)
- Rhys Parfitt (2016-2018, now Associate Professor, FSU)

Visiting Students and Scholars

@ UHM (since August 2024)

- Prof. Changhyun Yoo: Professor, Ewha Womans University, South Korea (2025–2026)

@ WHOI (until July 2024)

- Yoo-Jun Kim (University of Tokyo), Nov. 2023-Mar. 2024: High-resolution wind modeling.
- Peisen Tan (University of Miami), Summer of 2023: Laboratory and numerical modeling of short wind waves and swell interactions.
- Sam Bartusek (Princeton University, SSF 2019, now at Columbia University): Coastal Ocean impacts on landfalling ARs (Bartusek et al. 2021).
- Yuqing Liu (Bremen Univ., Guest Student, 2018, now at AWI, Germany): Submesoscale air-sea coupling
- Manthan Shah (Northeastern Univ., Guest Student, 2018): Scale-dependence of SST-wind coupling
- Xu Yuan (Univ. Twente, Guest Student, 2018): Upper-ocean variability in the tropical Indian Ocean (Yuan et al. 2020)
- Xiaolin Jin (Tsinghua University, Visiting Student, 2017), Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean. (Jin et al. 2018a, Jin et al. 2018b)
- Samuel Coakley (Rutgers University, SSF 2017): Southeast Asian monsoon variability in the CESM-Last Millennium Ensemble
- David Smith (Northeastern University, Guest Student, 2016): Decadal predictability of the North Atlantic Blocking
- Channing Prend (Columbia Univ., SSF 2016, Scripps Ph.D.): Impact of freshwater plumes on intraseasonal SST variability in the Bay of Bengal (Prend et al. 2018, Weller et al. 2019)
- Siraput Jongaramrungruang (Univ. Cambridge, SSF 2015, now at Caltech): Interannual modulation of intraseasonal convection in the Bay of Bengal Jongaramrungruang et al. 2017)
- Alicia Camacho (Valparaiso University, UCAR SOARS student 2015, Ph. D. from Stony Brook University): North Atlantic Oscillation, Jet and Blocking in CESM1 Large Ensemble Simulations. (Kwon et al. 2018)
- Carlos Martinez (Texas A&M University, UCAR SOARS student 2014, Ph.D. from Columbia University): North Atlantic Atmospheric Blocking and Atlantic Multi-decadal Oscillation in CESM1 Large Ensemble Simulations. (Kwon et al. 2018)

PROFESSIONAL ACTIVITIES

Community Activities

@ UHM (since August 2024)

- UCAR President's Advisory Committee on University Relations (PACUR): 2023-Present (link)
- Associate Editor, Journal of Climate (2019- Present)

@ WHOI (until July 2024)

- WHOI Representative for UCAR (2018-2023)
- Co-Chair, US CLIVAR Working Group on Mesoscale and Frontal-scale Ocean-Atmosphere Interactions and Influence on Large-scale Climate (2019-Present, link)
- Member, US CLIVAR Air-Sea Transition Zone Study Group (2022-2023, link)
- (Past) Member, AMS STAC Committee on Air-Sea Interaction (2019- 2022)
- (Past) Member, US CLIVAR Process Study and Model Improvement Panel (2016-2020)
- (Past) Associate Editor: Atmospheric Science Letters (2015-2017)

- (Past) Member, AMS STAC Committee on Coastal Environment (2011-2017)

Committee and Service

@ UHM (since August 2024)

- Department of Oceanography Awards Committee (2025- Present)

@ WHOI (until July 2024)

- WHOI Physical Oceanography Recruitment Committee for Scientific Staff (2020-2021)
- WHOI High-Performance Computing Strategy Committee (2022-2023)
- WHOI Summer Student Fellowship Coordinator (2017-2019)
- WHOI HPC User Advisory Committee (2010-2017)
- WHOI Physical Oceanography Department seminar series organizer (2010-2011)

Reviewer

Papers

- *Atmos. Sci. Lett.; Bull. Amer. Meteo. Soc.; Clim. Dynm; Comm. Earth & Env.; Env. Res. Lett.; Geo. Res. Lett.; J. Adv. Model. Earth Syst.; J. Atmos. Sci.; J. Climate; J. Mar. Syst.; J. Phys. Oceanogr.; J. Geophys. Res.-Atmospheres; J. Geophys. Res.- Oceans; Mon. Wea. Rev.; Nature; Nature Comms.; npj Climate and Atmos. Sci.; Ocean Dyn.; Ocean Modell.; Remote Sens.; Sci. Adv.; Sci. Rep,.etc.*

Proposals

- *NSF Physical Oceanography; NSF Climate and Large-Scale Dynamics; NSF Office of Polar Programs; NSF United States–Israel Binational Science Foundation; India Monsoon Missions I—III; Chile National Research Funding FONDECYT; CSCS Swiss National Supercomputing Centre*

Agency Proposal Review Panels

- NASA SWOT (2012); DOE ESMD (2020)

Society Memberships

- American Geophysical Union (2002-Present)
- American Meteorological Society (2002-Present)
- The Oceanography Society (2012-Present)

FUNDED RESEARCH (+ denotes Seo is the Lead or Sole PI)

@ UHM (since August 2024): Raised >$1.69M as lead PI

- +**ONR (current, $524,481):** Improving Air-Sea Flux Parameterizations to Better Understand the Diabatic Effects of Ocean and Surface Waves in Atmospheric Rivers.
- +**ONR (current, $632,333):** Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.
- +**WHOI (NASA) Subcontract (current, $314,183):** Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US.
- +**WHOI (DOE) Subcontract (current, $87,582):** Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas.
- +**WHOI (NSF) Subcontract (current, $139,693):** Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast.

hyodae@hawaii.edu https://www.HyodaeSeo.com

@ WHOI (until July 2024) External Grants: Raised ~$6.36M as lead PI, contributed to $11.49M as co-PI

- #**ONR (current, $50,000):** Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.
- #**NOAA (current, $773,753):** Exploiting coupled ocean-atmosphere-wave model simulations to identify observational requirements for air-sea interaction studies across the tropical Pacific. Co-PI: Wijffels
- #**NSF (current, $831,828):** Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast. Co-PI: Sauvage, Clayson, and Edson. ($139,693 subcontracted to UHM).
- #**NASA (current, $1,166,106):** Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US. ($314,183 subcontract to UHM).
- **DOE (current, $10M):** Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas. co-PI, PI: Kirincich (WHOI). ($87,582 subcontracted to UHM)
- #**NSF (current, $497,273):** Collaborative Research: Coupled Ocean-Atmosphere Feedbacks Affecting California Coastal Climate: Current Conditions and Future Projections.
- **NOAA (current, $658,442):** Regional multi-year prediction for the Northeast U.S. Continental Shelf. Co-PI. PI: Kwon (WHOI).
- #**NOAA (2019-2024, $767,746):** Coupled Ocean-atmosphere interaction mediated by ocean mesoscale eddies in the Northwestern Tropical Atlantic Ocean. Co-PI: Clayson (WHOI)
- #**ONR (2017-2023, $505,185):** Coupled Air-sea Interactions in the Bay of Bengal: Dynamics and Predictability of Monsoon Intraseasonal Oscillation.
- **NOAA (2017-2022, $267,797):** Upper ocean processes in the Maritime Continent and their impact on the air-sea interaction and MJO predictability, co-PI. Lead PI: Shinoda (TAMUCC)
- **NSF (2014-2019, $670,121):** Quantifying Predictability Limits, Uncertainties, Mechanisms, and Regional Impacts of Pacific Decadal Climate Variability, Institutional PI. Lead PI: Miller (SIO)
- **NOAA (2016-2019, $180,000):** Improvement of MJO simulation in NCEP Coupled Forecast System: Upper ocean and air-sea coupled processes, Institutional PI. Lead PI: Shionda (TAMUCC)
- **NSF (2014-2019, $500,000):** Decadal Variability in the North Atlantic Extra-Tropics: The Role of Coupling Between Atmospheric Blocking and the Atlantic Multidecadal Oscillation, co-PI. Lead PI: Ummenhofer (WHOI)
- #**ONR Young Investigator Award (2015-2019, $490,482):** Dynamics, Impacts, and Predictability of Coupled Ocean-Atmosphere Interactions in the Northern Indian Ocean.
- **NASA (2015-2018, $494,210):** Coupling Between the Atmospheric Intra-Seasonal Variability and Ocean Circulation in the Northern Hemisphere, co-I. PI: Kwon (WHOI)
- #**NSF (2014-2015, $72,000):** Regional Coupled Ocean-Atmosphere Feedback Processes Affecting Climate Along the California Coast.
- #**ONR (2011-2014, $91,000):** Coupled Ocean-Atmosphere Dynamics and Predictability of MJO's.

@ WHOI (until July 2024) Internal (competitive) Grants, Secured ~$495K

- #**Fowler Ocean and Climate Fellow (2021-2023, $140,213):** Western boundary currents and air-sea interactions: toward skillful predictive understanding and modeling of extreme storm and short-term climate events
- #**Independent Study Award (2019-2021, $64,000):** Southern Ocean Eddies and Atmospheric Storm Tracks
- #**Ocean and Climate Change Institute (2015-2017, $55,000):** A regional coupled model with explicit convection: diurnal rainfall and air-sea-land interaction in the Maritime Continent.
- #Independent Study Award (2013-2015, $49,000): **Submesoscale air-sea Interactions.**
- #**Ocean and Climate Change Institute (2011-2013, $140,000):** Dynamical analysis of surface wind responses to sea ice and surface temperature variations in the Arctic Ocean: Synthesis of data and model simulation. Co-PI: Yang (WHOI)
- #**Independent Study Award (2012-2013, $47,000): T**rends in coastal upwelling and sea breeze in the California-Oregon Coast: Land-ocean-atmosphere interactions in data and model.